Indiana Insurance & General Chat Assistant

Category: CAT3

1. Project Overview

This project implements a local retrieval-augmented chatbot using Streamlit, LangChain, and Ollama. The assistant can answer both Indiana insurance–related and general questions. When the query concerns insurance, it retrieves context from a Chroma vector database built from Indiana Department of Insurance documents, grounding its response in verified information.

2. Idea Justification

I did not switch ideas during the project. My original goal was to build a retrieval-augmented local chatbot capable of handling insurance-related inquiries in America with structured, factual answers. But due to technical limitations I can only do insurance-related inquiries in Indiana because I can not have a 10T dataset. Thus, this chatbot project continues the same design philosophy, applied now in a real-world RAG context.

3. Prompt Engineering Techniques and Design Process

(a) Role-Playing Prompt (Lines 47–49)

The system prompt explicitly defines the assistant's identity, goal, and ethical scope:

> "You are an Indiana consumer and insurance assistant… You are NOT a lawyer…"

(b) Varying Generation Parameters (Lines 26–38)

I implemented dynamic temperature adjustment in the `get_llm()` function:

- `strict = 0.1` → logical
- `normal = 0.4` → balanced
- `creative = 0.8` → flexible

It allows users to trade off between factual precision and expressiveness while using the same base model.

(c) Task (Ask) Decomposition (Line 52)

In the system prompt, I instruct the model to:

> "First, briefly break the user question into 2–3 subtasks or key issues."

(d) Multi-Path Reasoning and Structured Outputs (Lines 66–157)

From line 66 onward, the code branches depending on query type:

- Insurance questions use a complex, structured prompt (`build_insurance_prompt`)
- General questions use a simpler conversational prompt (`build_general_prompt`).

The structured output format makes sure if it is about insurance it will have a more advanced answer as asked in the assignment requirements.

(e) Rationale Behind Technique Selection

I reused these four techniques—ask decomposition, multi-path reasoning, structured outputs, and role-playing—because in my HW2 experiments with the same model, they produced the highest reasoning accuracy and response coherence across tasks. And I added a varying generation parameters technique for choosing temperature just because I think it is fun to try something new and it should not affect accuracy.